OS & SP

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SKILL – 03

1.

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GNU nano 8.7.1

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_HISTORY 5
#define MAX_LENGTH 100

int main() {
    char history[MAX_HISTORY][MAX_LENGTH];
    char input[MAX_LENGTH];
    int count = 0;
    int current = -1;

    printf("=== Command History Program ===\n");
    printf("Enter commands (type 'exit' to stop)\n\n");

    while (1) {
        printf("Command: ");
        fgets(input, MAX_LENGTH, stdin);

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        if (count < MAX_HISTORY) {
            strcpy(history[count], input);
            count++;
        } else {
            for (int i = 1; i < MAX_HISTORY; i++)
                strcpy(history[i - 1], history[i]);

            strcpy(history[MAX_HISTORY - 1], input);
        }

        current = count;

        printf("Command stored successfully.\n");
    }

    printf("\n=== Command History ===\n");

    for (int i = 0; i < count; i++) {
        printf("%d: %s\n", i + 1, history[i]);
    }

    printf("\nEscape Sequence Demonstration:\n");
    printf("New Line -> Hello\nWorld\n");
    printf("Tab -> Hello\tWorld\n");
    printf("Backslash -> C:\\Program Files\n");

    return 0;
}
```

Output :

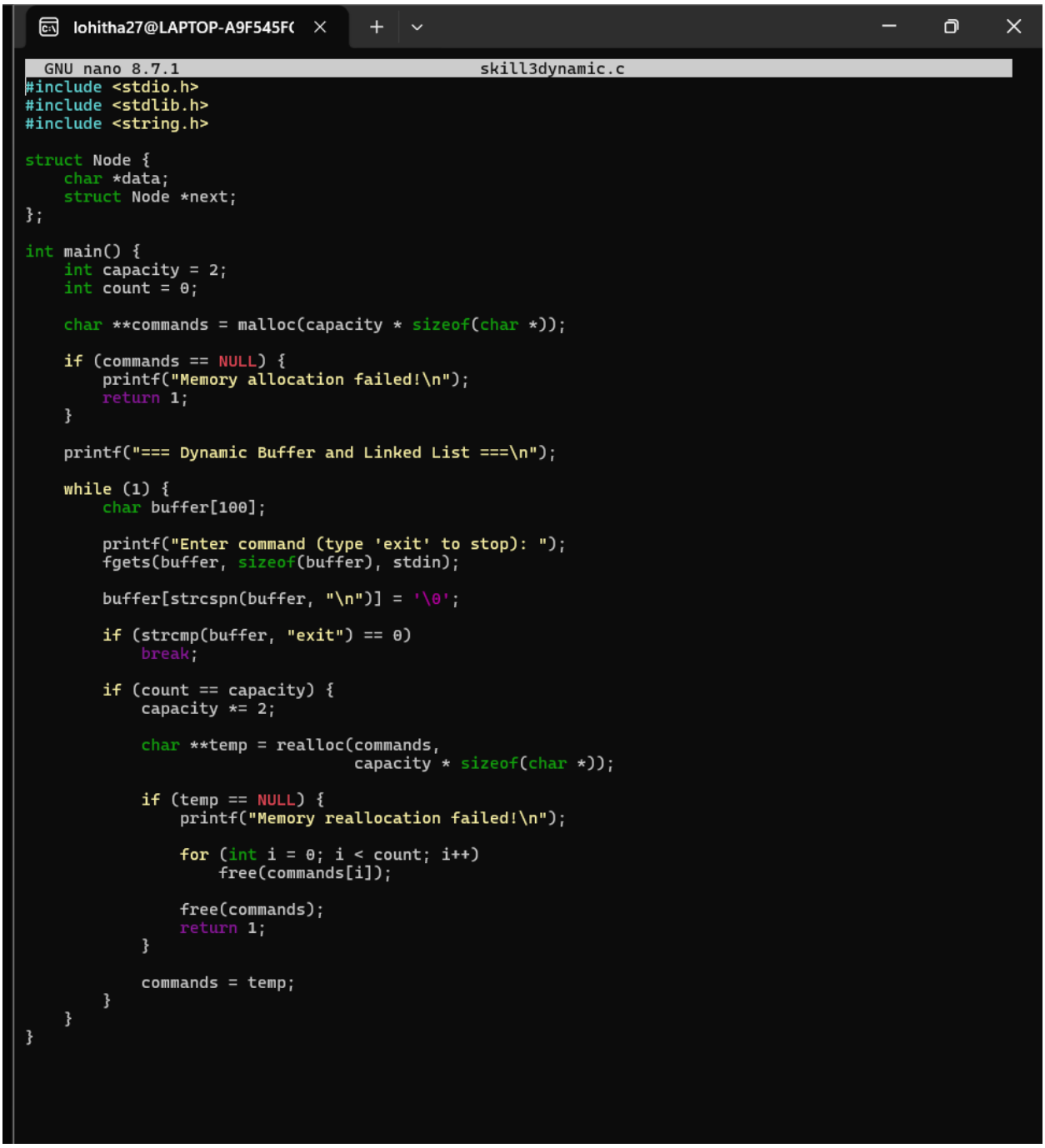
```
skill3.c:9:18: note: each undeclared identifier is reported only once for each function it appears in
lohitha27@LAPTOP-A9F545FG:~/SKILL$ nano skill3.c
lohitha27@LAPTOP-A9F545FG:~/SKILL$ gcc skill3.c -o skill3
lohitha27@LAPTOP-A9F545FG:~/SKILL$ ./skill3
=== Command History Program ===
Enter commands (type 'exit' to stop)

Command: ls
Command stored successfully.
Command: pwd
Command stored successfully.
Command: history
Command stored successfully.
Command: exit

=== Command History ===
1: ls
2: pwd
3: history

Escape Sequence Demonstration:
New Line -> Hello
World
Tab -> Hello    World
Backslash -> C:\Program Files
lohitha27@LAPTOP-A9F545FG:~/SKILL$
```

2.


lohitha27@LAPTOP-A9F545F(
GNU nano 8.7.1 skill3dynamic.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
struct Node {
char *data;
struct Node *next;
};
int main() {
int capacity = 2;
int count = 0;
char **commands = malloc(capacity * sizeof(char *));
if (commands == NULL) {
printf("Memory allocation failed!\n");
return 1;
}
printf("=== Dynamic Buffer and Linked List ===\n");
while (1) {
char buffer[100];
printf("Enter command (type 'exit' to stop): ");
fgets(buffer, sizeof(buffer), stdin);
buffer[strcspn(buffer, "\n")] = '\0';
if (strcmp(buffer, "exit") == 0)
break;
if (count == capacity) {
capacity *= 2;
char **temp = realloc(commands,
capacity * sizeof(char *));
if (temp == NULL) {
printf("Memory reallocation failed!\n");
for (int i = 0; i < count; i++)
free(commands[i]);
free(commands);
return 1;
}
commands = temp;
}
}
}

```
lohitha27@LAPTOP-A9F545FG:~/SKILL$ nano skill3dynamic.c
lohitha27@LAPTOP-A9F545FG:~/SKILL$ gcc skill3dynamic.c -o skill3dynamic
lohitha27@LAPTOP-A9F545FG:~/SKILL$ ./skill3dymanic
-bash: ./skill3dymanic: No such file or directory
lohitha27@LAPTOP-A9F545FG:~/SKILL$ nano skill3dynamic.c
lohitha27@LAPTOP-A9F545FG:~/SKILL$ gcc skill3dynamic.c -o skill3dynamic
lohitha27@LAPTOP-A9F545FG:~/SKILL$ ./skill3dynamic
=== Dynamic Buffer and Linked List ===
Enter command (type 'exit' to stop): ls
Enter command (type 'exit' to stop): pwd
Enter command (type 'exit' to stop): gcc test.c
Enter command (type 'exit' to stop): exit
lohitha27@LAPTOP-A9F545FG:~/SKILL$
```